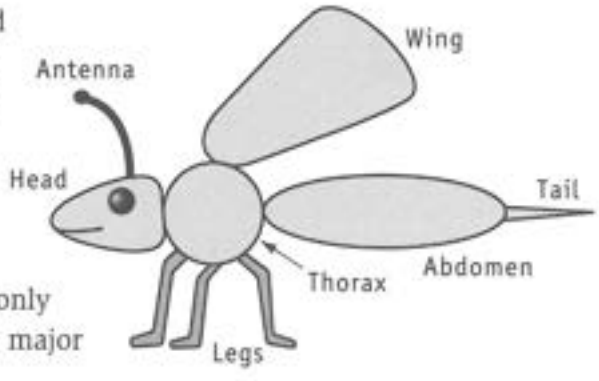
Variations in a Species

23A

You will need scissors, glue and a die to do this exercise.

Some entomologists studied a species of insect only found on Gribble Island. These unique insects seemed to have many differences, but were definitely of the same species because they could breed with each other.

The diagram shows a typical specimen found on Gribble Island.



After a lot of research it was discovered that there were only two phenotypes (physical characteristics) for each of the major body parts of the insects.

You can simulate the range of characteristics of these insects by following these instructions:

- collect sheet 23B showing possible body parts ('Insect parts'), glue, scissors and a die.
- roll the die to randomly select the characteristics of a body part. Use the table below to make your selection, depending on whether the throw of the die results in an even or odd number. The first roll of the die will select the type of head, the second will select the antenna, etc. Some characteristics (marked *) will require two throws of the die.
- assemble the insect parts, and glue the complete insect into your book. Label the parts.

Characteristic	Even Number	Odd Number
Head	Plain eye	Compound eye
Antenna	Long antenna	Short antenna
Thorax	Striped thorax	Plain thorax
Legs	Long legs	Short legs
Wings*	Large wings	Small wings
	Checkered wings	Plain wings
Abdomen*	Thick abdomen	Thin abdomen
	Dark abdomen	Light abdomen
Tail	Straight tail	Curly tail

Further Activities

1. After completing your insect, compare it with others made in the class.
2. Explain why it is unlikely that any two people in the class will have made identical creatures.
3. A scientist found a group of insects on another island that resembled those from Gribble Island, but did not breed with them. Are these insects of the same species as the ones you have been studying? Explain your answer.

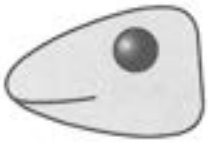
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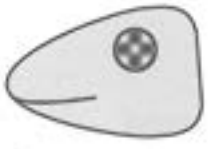
Variations in a Species

23B

Insect parts



Plain eye



Compound eye



Long antenna



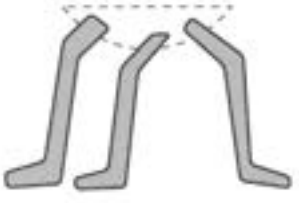
Short antenna



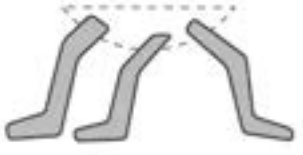
Plain thorax



Striped thorax



Long legs



Short legs

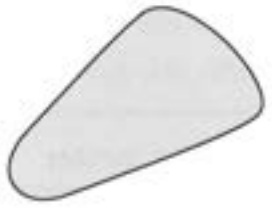


Plain



Chequered

Large wings



Plain



Chequered

Small wings

Dark



Light



Thick abdomens

Dark



Light



Thin abdomens



Straight tail



Curly tail

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